II – SOME KEY PROBABILITY CONCEPTS AND LAWS

(1) Conditional probability

The **conditional** probability of an event A, given that an event B has occurred, is equal to

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$ provided that P(B>0)

This is a key concept in statistics. Along with conditional expectation, it provides the basic conceptual underpinnings of regression analysis. More on this coming later in the course.

(2) Independence of Events

If learning that B has occurred does not change the probability of A, then we say that A and B are independent.

Two events are **independent** if

$$P(A \cap B) = P(A)P(B)$$

If P(A)>0 and P(B)>0 then A and B are independent if and only if:

P(A|B)=P(A)
P(B|A)=P(B)

The Multiplicative Law of Probability:

$$P(A \cap B) = P(A)P(B \mid A)$$
$$P(A \cap B) = P(B)P(A \mid B)$$

The Additive Law of Probability:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Law of Total Probability

Assume that $\{ A_1, A_2, ..., A_k \}$ is a partition of S such that $P(A_i) > 0$, for i=1,2,…, k. Then for any event B , $P(B) = \sum_{i=1}^{k} P(B \mid A_i)P(A_i)$

Special case: $P(B) = P(B \mid A)P(A) + P(B \mid \overline{A})P(\overline{A})$ where $\overline{A}$ is the complement of A

Ex: $P(College) = P(College \mid Female)P(Female) + P(College \mid Male)P(Male)$

9 PROBABILITY DISTRIBUTIONS

(WMS: Chapter 2-4)

9.1 Probability of Events

We talk about the probability of an event, which is the outcome of an experiment. The sample space consists of all simple events which cannot be decomposed into simpler events. An event, E, is a subset of a sample space, S.

The **probability** $P(E)$ of an event E is a number representing the chance that the event occurs. It is defined such that

- $0 \leq P(E) \leq 1$
- $P(S) = 1$
- $P(\overline{E}) = 1 - P(E)$, where $\overline{E}$ (or E^c) is the complement of E
- If E_i, E_j are pairwise mutually exclusive, so $E_i \cap E_j = \emptyset$(for $i \neq j$), then
$$P(E_1 \cup E_2 \cdots) = \sum_{i=1}^{\infty} P(E_i).$$

Conditional Probability
For events A, B, we define:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ if } P(B) > 0$$

$P(A|B)$ represents the probability of A given B; that is, the relative frequency of A given B. So

$$P(A \cap B) = P(A|B) \cdot P(B)$$

Independence
Events A, B are independent if any one of the following holds:

- $P(A|B) = P(A)$; that is, knowing B has happened doesn't tell you anything about the likelihood of A happening
- $P(B|A) = P(B)$; that is, knowing A has happened doesn't tell you anything about the likelihood of B happening
- $P(A \cap B) = P(A) \cdot P(B)$; that is, the same fraction of As happen with B as in whole population

Ex. P(death in auto accident) and P(death auto accident | wearing seat belt) are not equal. These events (wearing seat belt and death in auto accident) are not independent.

Ex. To investigate racial profiling, which of the following do you want to compare: P(White|Frisk), P(Black|Frisk), P(Frisk|White), P(Frisk|Black)?

Solution: We want to compare the probability of being frisked for Whites and Blacks; that is P(Frisk|White), P(Frisk|Black).

9.2 Random Variables and Probability Distributions

A **random variable**, Y, is a real-valued function whose domain is the sample space. In other words, Y is a function whose numerical value depends on outcome of an experiment.

There are two types of random variable. A **discrete** random variable takes on discrete or countably infinite values. Otherwise, Y is a **continuous** random variable.

ex. Number of days Dow Jones average is above a specific value in the current year: Discrete.
ex. Length of time Dow Jones average is above a specific value in the current year: Continuous.

To represent the probabilities of all the possible values of Y, we use a **distribution**. Distributions are defined differently for discrete and continuous variables so we treat them separately.

9.2.1 For Discrete Random Variable

Probability Distribution Function (p.d.f.), $p(y)$, defined by

$$p(y) = P(Y = y).$$

See Figure 1, showing the relative frequency of y. Properties of p:

- $0 \leq p(y) \leq 1$
- $\sum_y p(y) = 1$.

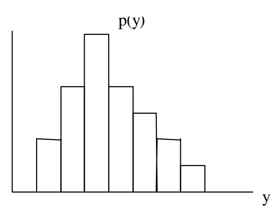

Figure 1: Horizontal axis is y-values. Vertical axis is relative frequency

Cumulative Distribution Function (c.d.f.), $F(y)$ defined by

$$F(y) = P(Y \leq y)$$

See Figure 2. Properties of F:

- $\lim_{y\to-\infty} F(y) = 0, \lim_{y\to\infty} F(y) = 1,$
- $F(y)$ is increasing so $y_1 < y_2 \Rightarrow F(y_1) \leq F(y_2)$.

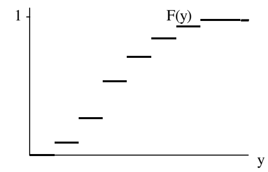

Figure 2:

ex. For three tosses of a fair coin, let Y be the total number of heads. The p.d.f. and c.d.f. of Y are:

y	0	1	2	3
$p(y)$	1/8	3/8	3/8	1/8
$F(y)$	1/8	4/8	7/8	1

For a Continuous Random Variable:
If Y can take an uncountable number of values, then the probability of getting any specific value, y, is zero; that is $P(Y = y) = 0$. Thus we can't use the same kind of probability distribution function as we used for a discrete random variable. However, the cumulative distribution function is defined as before, so we start with this.

Cumulative Distribution Function (c.d.f.), F, is defined by
$F(y) = P(Y \leq y)$ gives probability that $Y \leq y$. See Figure 3. Properties of F:

- $\lim_{y\to-\infty} F(y) = 0$, $\lim_{y\to\infty} F(y) = 1$,
- $F(y)$ increasing.

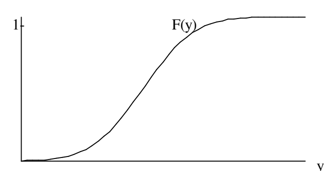

Figure 3:

Probability Density Function (p.d.f.), f,
Assuming that the derivative exists, we define $f(y) = F'(y)$. See Figure 4.
Properties of f:

- $f(y) \geq 0$
- $\int_{-\infty}^{\infty} f(y)dy = 1$

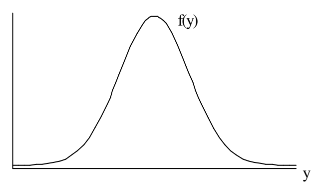

Figure 4:

To Calculate Probabilities for Continuous Random Variables:

$$P(a \leq Y \leq b) = P(Y \leq b) - P(Y < a) = F(b) - F(a).$$

Since $f(y) = F^{'}(y)$, by the Fundamental Theorem of Calculus, we have

$$P(a \leq Y \leq b) = F(b) - F(a) = \int_a^b f(y)dy.$$

See Figure 5.

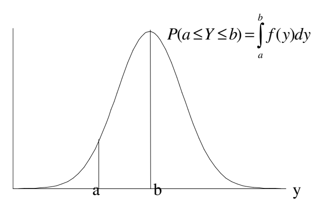

Figure 5:

What does the pdf tell you?
Values of $f(y)$ don't tell you anything directly. If $f(10) = 0.15$, probability that $Y = 10$ is not 0.15. The value of f tells you that the probability that Y is within $10 \leq y \leq 10 + dy$ is $0.15dy$. In other words,

$$P(y \leq y \leq y + dy) = f(y)dy$$

Thus a large value of $f(y)$ tells you that values near y are very likely. This is why f is called a density function.